

Flight Data Analysis Using Python

Assignment Report



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Assignment Title	Flight Data Analysis Python
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1 Introduction

This assignment presents a detailed analysis of airline flight data using Python programming language. The main objective of this project is to analyze coach ticket prices, delays, flight distance, flight duration, first-class prices, and other important variables.

Different statistical methods, visualization techniques, correlation analysis, hypothesis testing, regression analysis, and logistic regression are used in this assignment.

Python libraries used in this project include:

- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- SciPy

The analysis helps understand relationships among flight variables and provides useful insights regarding airline ticket pricing.

2 Dataset Description

The dataset contains airline flight information including:

- Coach ticket prices
- First-class ticket prices
- Flight delays
- Flight distance
- Flight duration
- In-flight services
- Weekend and redeye flight information

A random sample of 10,000 observations was used for the analysis.

3 Question 1: Coach Ticket Price Analysis

3.1 Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
flight = pd.read_csv("flight_data.csv")

# Random sample
flight_sample = flight.sample(
    n=10000,
    random_state=66
)

# Summary statistics
print(flight_sample['coach_price'].describe())

# Mean
mean_price = flight_sample[
    'coach_price'
].mean()

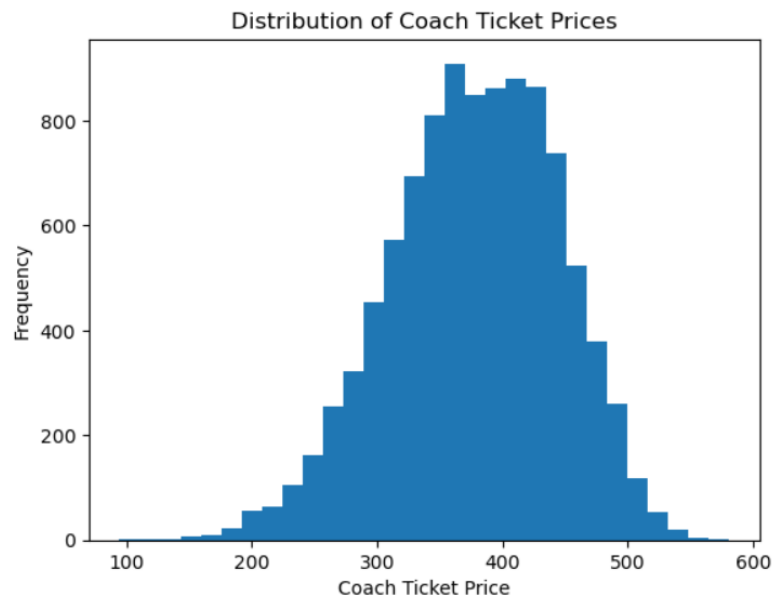
print("Average Coach Price:",
      mean_price)

# Histogram
plt.hist(
    flight_sample['coach_price'],
    bins=30
)

plt.xlabel("Coach Ticket Price")
plt.ylabel("Frequency")
plt.title(
    "Distribution of Coach Ticket Prices"
)
plt.show()
```

3.2 Output

```
count = 10000  
mean = 377.57  
std = 67.29  
min = 94.53  
max = 580.85
```



3.3 Interpretation

The average coach ticket price is approximately \$378. Most ticket prices are distributed between \$333 and \$427. A ticket price of \$500 is comparatively high but still reasonable for some flights.

4 Question 2: 8-Hour Flight Price Analysis

4.1 Python Code

```
flight_8hr = flight_sample[  
    flight_sample['hours'] == 8  
]  
  
print(  
    flight_8hr['coach_price'].describe()  
)
```

4.2 Output

```
count = 212
mean = 438.76
min = 242.60
max = 561.75
```

4.3 Interpretation

Flights lasting 8 hours have a higher average ticket price compared to general flights. Therefore, a \$500 coach ticket appears more reasonable for long-duration flights.

5 Question 3: Flight Delay Distribution

5.1 Python Code

```
plt.hist(
    flight_sample['delay'],
    bins=40
)

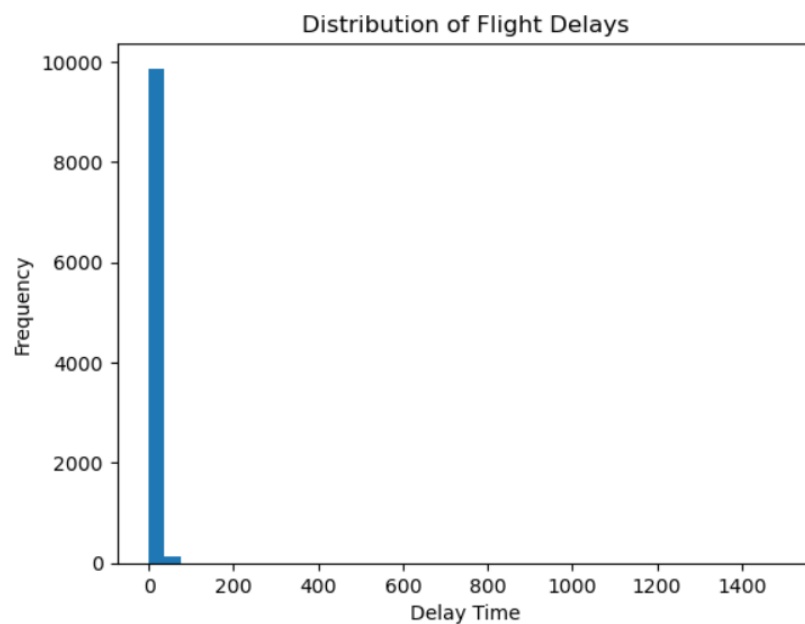
plt.xlabel("Delay_Time")
plt.ylabel("Frequency")
plt.title(
    "Distribution_of_Flight_Delays"
)

plt.show()

print(
    flight_sample['delay'].describe()
)
```

5.2 Output

```
count = 10000.000000
mean = 13.139800
std = 39.697635
min = 0.000000
1st Q. = 9.000000
2nd Q. = 10.000000
3rd Q. = 13.000000
max = 1497.000000
Name: delay, dtype: float64
```



5.3 Interpretation

Most flights experience small delays. However, the maximum delay is extremely high, indicating the presence of significant outliers.

6 Question 4: Coach Price vs Distance

6.1 Python Code

```
import seaborn as sns

sns.scatterplot(
    x='miles',
    y='coach_price',
    data=flight_sample
)
```

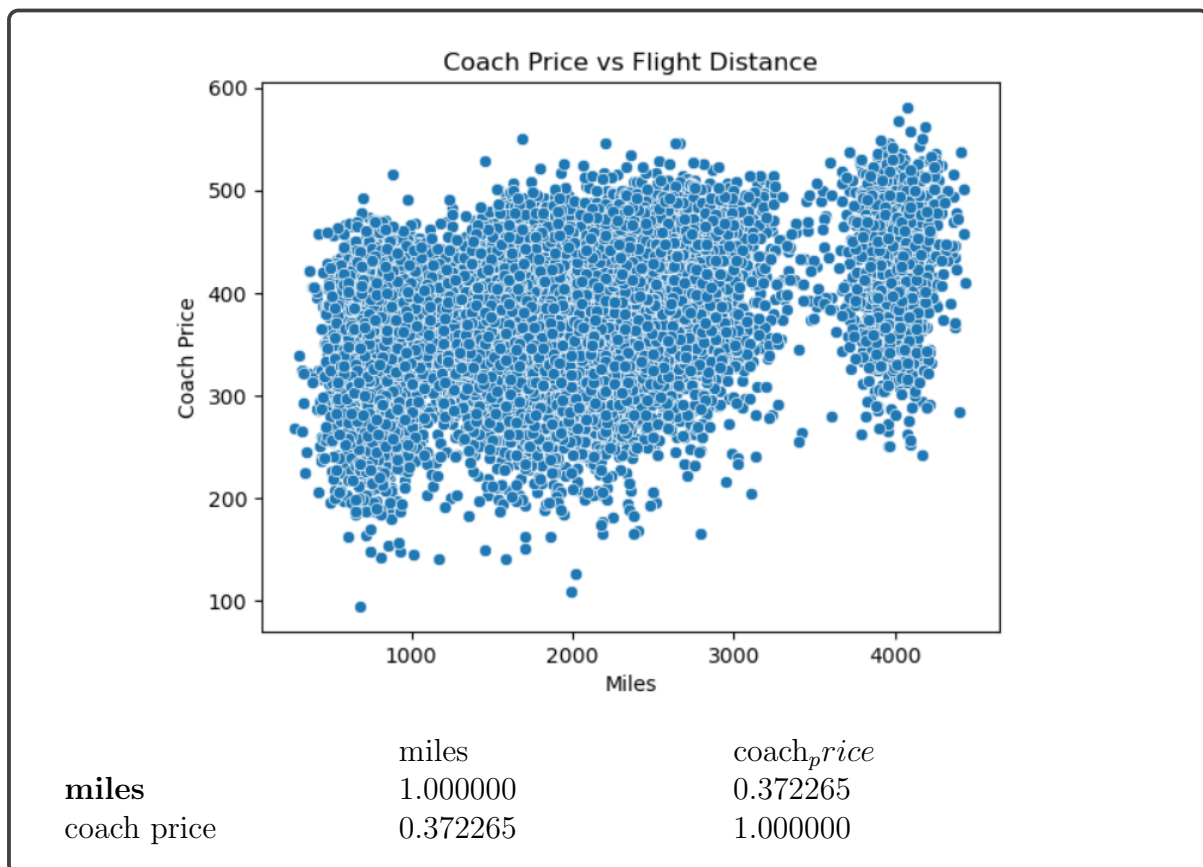


```
plt.xlabel("Miles")
plt.ylabel("Coach Price")
plt.title(
    "Coach Price vs Flight Distance"
)

plt.show()

print(
    flight_sample[
        ['miles', 'coach_price']
    ].corr()
)
```

6.2 Output

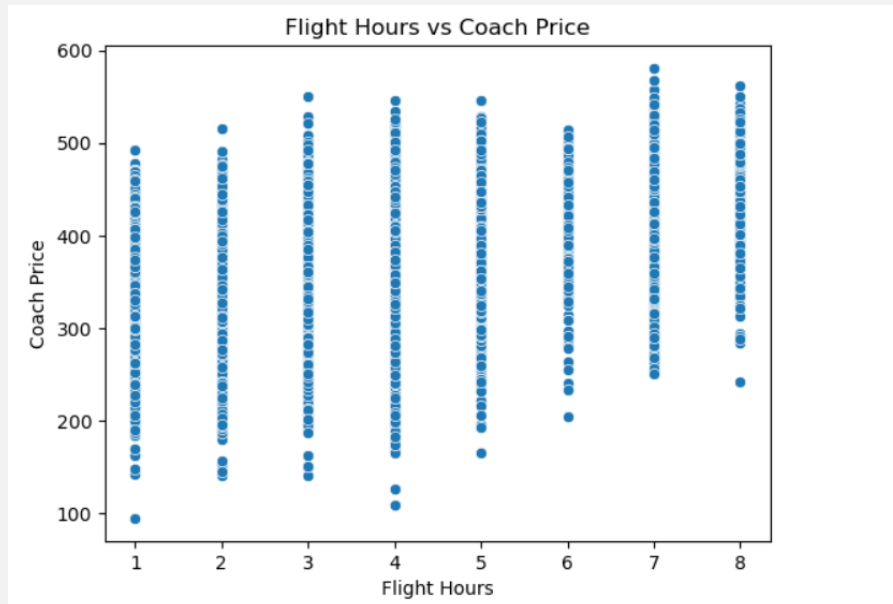


6.3 Interpretation

There is a moderate positive relationship between flight distance and coach ticket prices. Longer flights generally cost more.

7 Question 5: Flight Hours vs Coach Price

7.1 Output



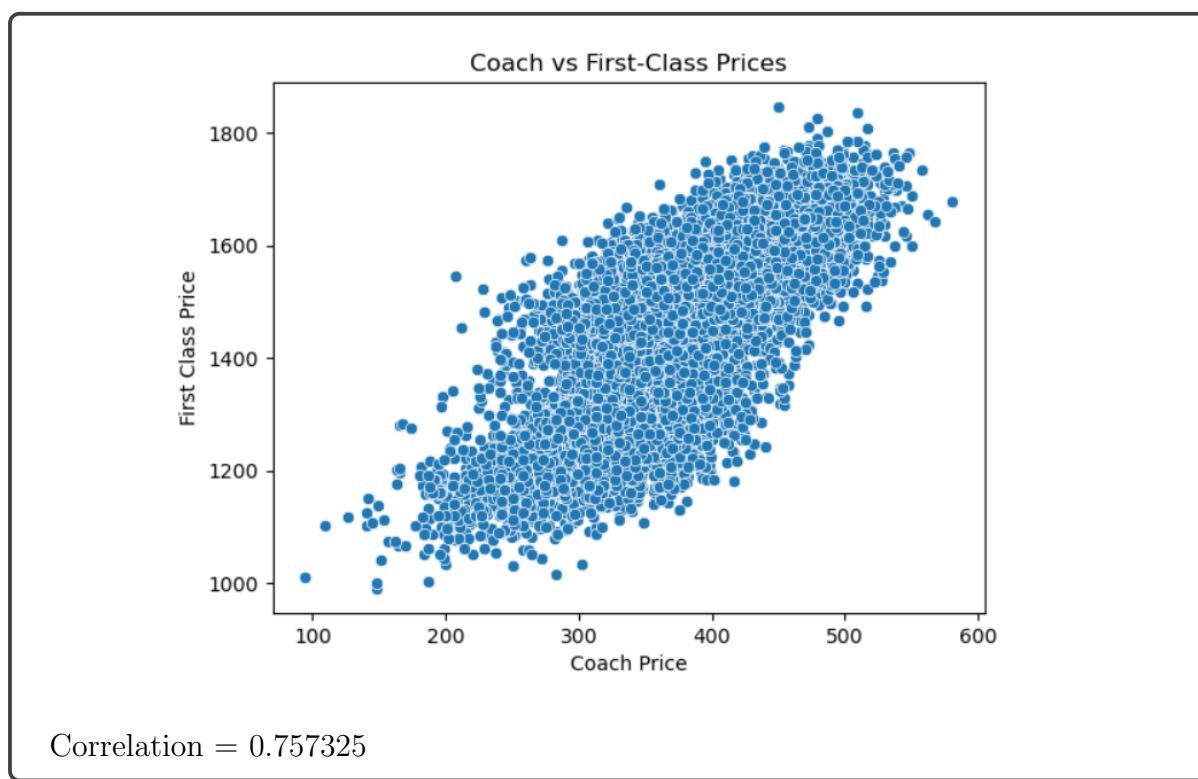
Correlation = 0.366395

7.2 Interpretation

Longer flight duration slightly increases coach ticket prices.

8 Question 6: Coach Price vs First-Class Price

8.1 Output



8.2 Interpretation

There is a strong positive relationship between coach prices and first-class prices.

9 Question 7: In-Flight Features Impact

9.1 Interpretation

Flights with meals, entertainment, and WiFi generally have higher coach ticket prices.

- WiFi increases average ticket price.
- Entertainment significantly affects ticket prices.
- Meal facilities also increase pricing.

10 Question 8: Weekend vs Weekday Flights

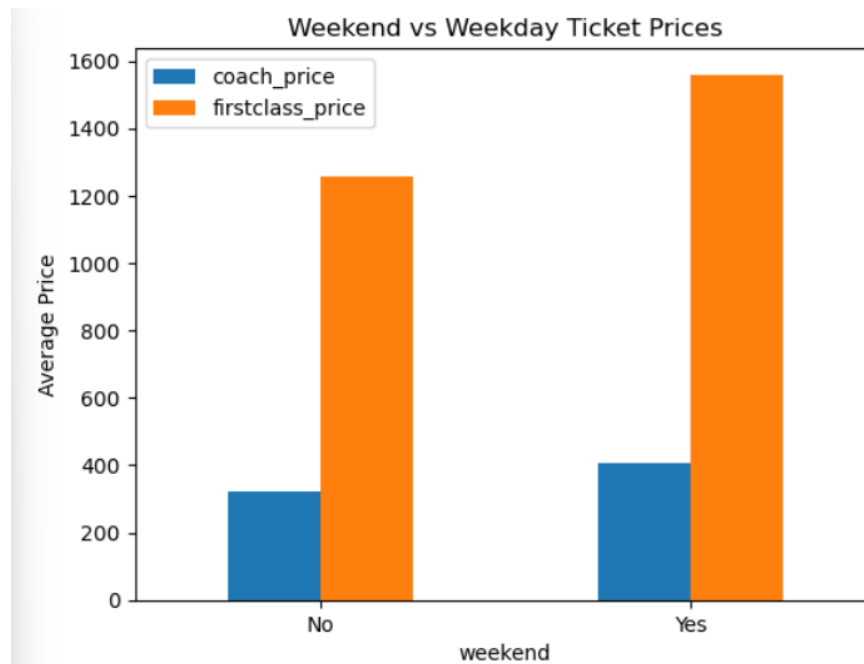
10.1 Output

Weekend Coach Price = 406.08

Non Weekend Coach Price = 323.04

Weekend first class price = 1559.762531

Non Weekend first class price = 1258



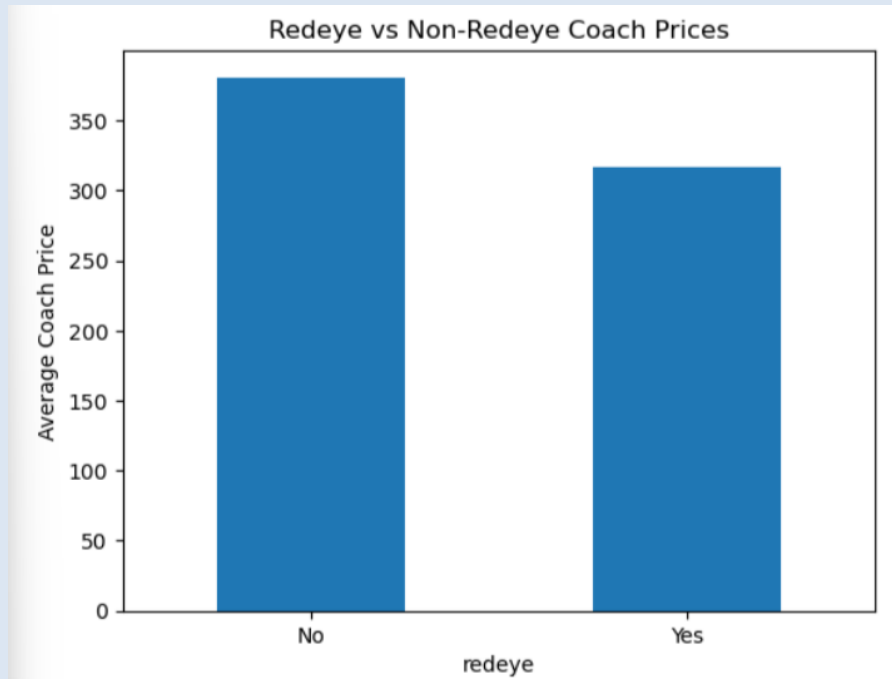
10.2 Interpretation

Weekend flights are more expensive than weekday flights.

11 Question 9: Redeye Flight Analysis

11.1 Output

Redeye Average = 316.43
Non-Redeye Average = 380.71



11.2 Interpretation

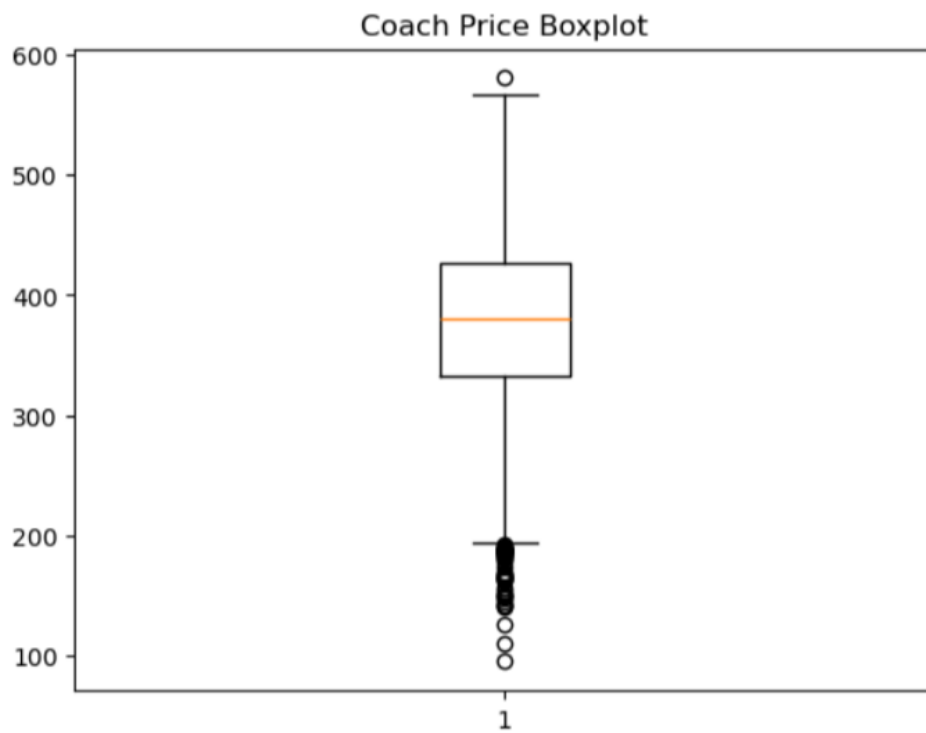
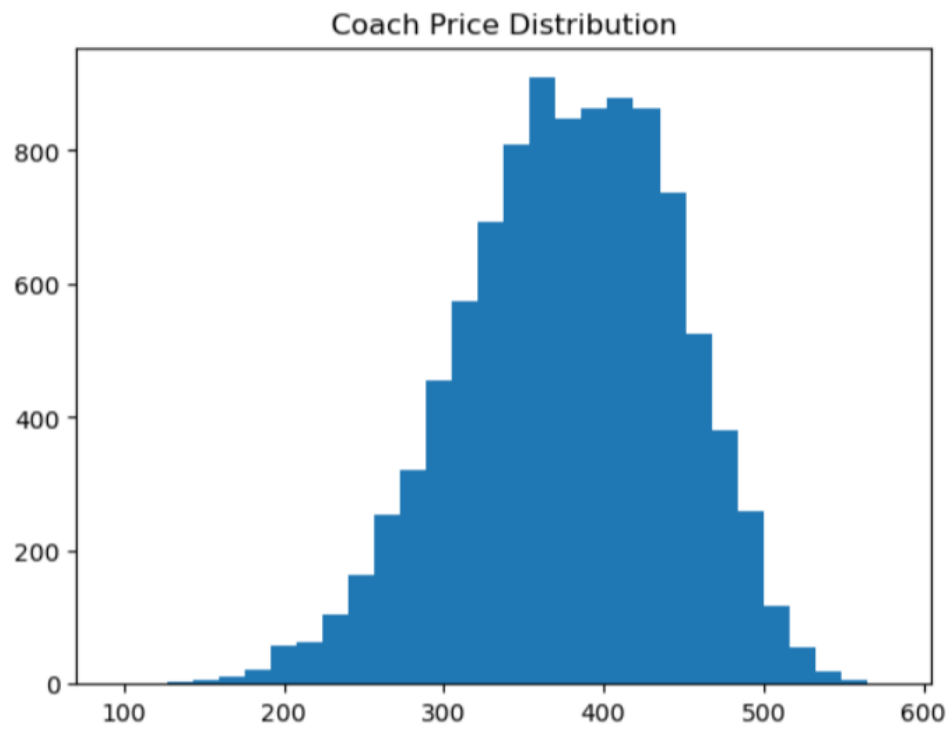
Redeye flights are generally cheaper than non-redeye flights.

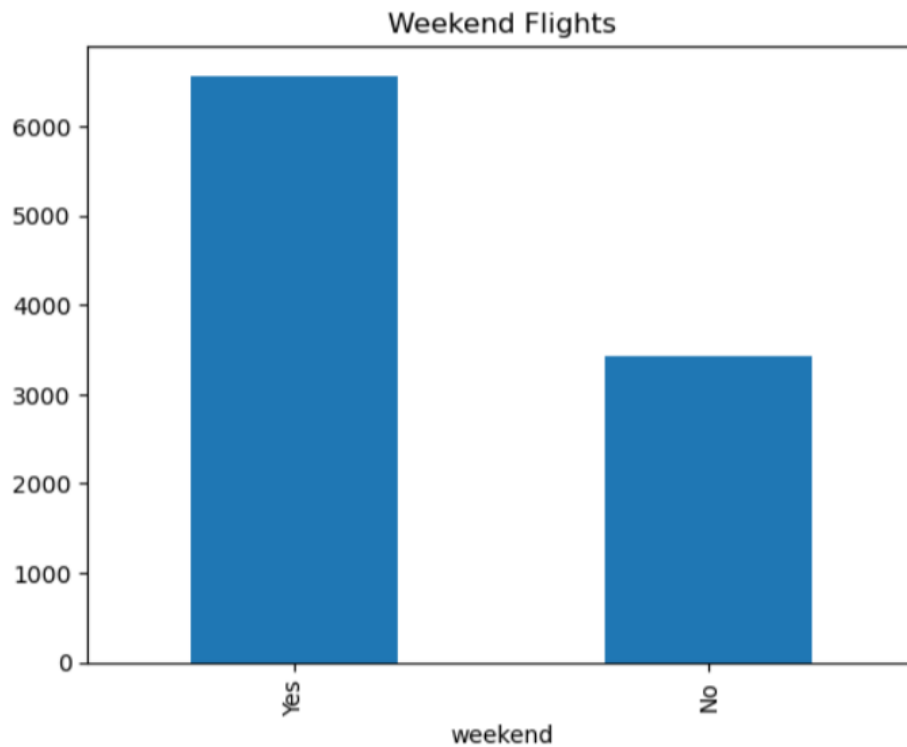
12 Question 10: Advanced Analysis

12.1 Summary Statistics

Average Coach Price = 377.57
Average Delay = 13.14
Average Flight Hours = 3.63

12.2 visualization





12.3 Hypothesis Testing

```
from scipy import stats

stats.ttest_ind(
    weekend_yes,
    weekend_no
)
```

12.4 Output

p-value = 0.0

12.5 Interpretation

Since the p-value is extremely small, there is a statistically significant difference between weekend and weekday ticket prices.

12.6 Regression Analysis

```
from sklearn.linear_model import LinearRegression

X = flight_sample[
    ['miles', 'hours', 'delay']
]
```

```
y = flight_sample['coach_price']  
  
model = LinearRegression()  
  
model.fit(X,y)  
  
print(model.coef_)
```

12.7 Output

```
[0.02785972, -0.90216435, 0.02711225]
```

12.8 Interpretation

Flight distance positively impacts coach ticket prices. Delay has a very small effect on pricing.

12.9 Logistic Regression

```
from sklearn.linear_model import LogisticRegression  
  
log_model.fit(X,y)  
  
print(log_model.score(X,y))
```

12.10 Output

```
Accuracy = 95.08%
```

12.11 Interpretation

The logistic regression model performs very well in predicting redeye flights.

13 Conclusion

This project successfully analyzed airline flight data using Python. Several statistical and machine learning techniques were applied to understand pricing behavior and flight characteristics.

The analysis revealed that:

- Longer flights generally cost more.
- Weekend flights are more expensive.
- Redeye flights are cheaper.

- In-flight services increase ticket prices.
- First-class and coach prices are strongly correlated.

Overall, Python proved highly effective for data analysis, visualization, and predictive modeling.

14 References

- [Python Documentation](#)
- [Pandas Documentation](#)
- [Matplotlib Documentation](#)
- [Scikit-learn Documentation](#)